

ME5412 Robotics for Healthcare

Interfaces for Assistive Technologies

辅助技术交互接口

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Outline of the Lecture

- Elements of AT interfaces
- 辅助技术接口的组成要素
- Characteristics of AT interfaces
- 辅助技术接口的特性
- Adapted interfaces: keyboards, voice recognition, touch screens, eye-gaze systems, pointing devices
- 适配型接口：键盘、语音识别、触摸屏、眼动追踪系统、定点输入设备
- Indirect interfaces
- 间接式接口
- User input and human skills
- 用户输入与人体操作能力
- Interfaces for indirect selection: mechanical switches and other switches
- 间接选择所用接口：机械开关与其他类型开关

Assistive Technology (AT)

Any AT device consists of 4 main components:

任何辅助技术设备都包含4个主要组成部分：

1. Human-AT interface (HTI): it can be physical or cognitive, it is bi-directional

人-辅助技术接口（HTI）：可以是物理层面或者认知层面，双向交互

2. Activity output: it contributes to the functional performance

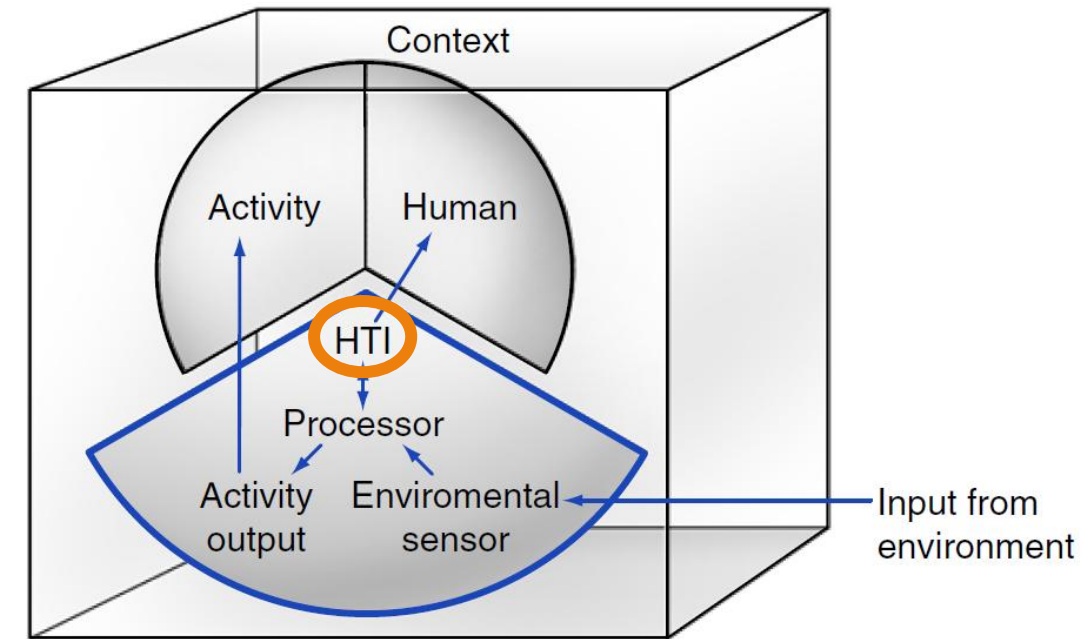
行为输出：用来实现功能执行

3. Processor: it translates physical (forces) and cognitive (information) inputs into signals used to control the activity output

处理器：把物理输入（力）、认知输入（信息）转换为信号，用来控制行为输出

4. Environmental sensor: it detects information on the context and transfers them to the processing unit and to the human, through the human-AT interface

环境传感器：感知环境上下文信息，将信息传给处理单元，并通过人-辅助技术接口反馈给使用者。



主流 vs. 适配型 人-辅助技术接口

Mainstream vs. Adapted HTIs

- 适配型人-辅助技术接口主要供给精细运动控制能力受损的人群；这类人群难以使用普通接口：电脑键盘、鼠标/触摸屏这类定点设备、电灯开关、电视机等娱乐设备按键，以及其他日常用品。
- Adapted HTIs are typically used by individuals who have **reduced (fine) motor control** that makes it difficult to use common HTIs such as computer keyboards and pointing devices (e.g., a mouse, or touch screen); light switches; controls on entertainment devices such as TVs; and similar everyday items
- Adapted HTIs are also useful to people who need to use a powered wheelchair or use adapted controls to drive a vehicle • 适配型接口同样适用于电动轮椅使用者，或是需要改装控制器来驾驶交通工具的人群。

Reduced motor control may depend on: 运动控制能力下降的诱因：

- Muscle tone (high or low), reduced strength, reduced endurance, reduced range of motion, presence of tremor and the type of tremor • 肌张力（过高/过低）、肌力下降、耐力不足、关节活动范围受限、存在震颤以及震颤的类型。
- Neuromotor impairment, (e.g., CP, TBI, ALS) in which there is an impairment of motor control due to abnormalities of tone, • 神经运动功能损伤（例如脑瘫CP、脑外伤TBI、肌萎缩侧索硬化ALS），运动控制受损来源于肌张力异常
- SCI where there is limited/absent motor function, • 脊髓损伤SCI：运动功能受限或完全丧失，
- MD and other dystrophies where there is low endurance and limited ROM, • 肌营养不良MD及其他肌营养不良病症：耐力差、关节活动范围受限，
- Musculoskeletal diseases (e.g., RA) that also affects ROM, endurance, and strength 肌肉骨骼疾病（例如类风湿关节炎RA），同样会影响关节活动范围、耐力与肌力。

做了适配改造的标准键盘

Standard Keyboards with Adaptations

- Keyboard shields,
reducing accidental entries

- 键盘防护挡板：防止误触输入



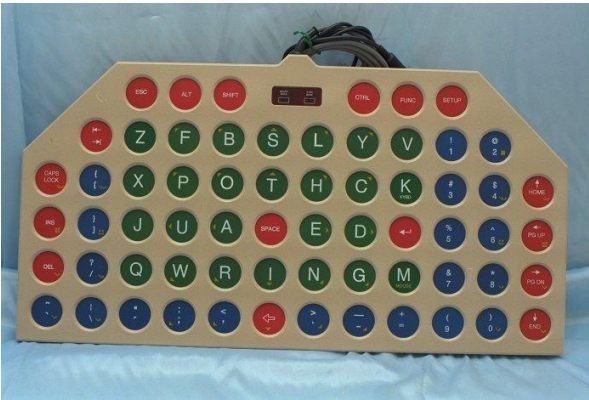
- Ergonomic keyboards

- 人体工学键盘



替代键盘（直接选择式）

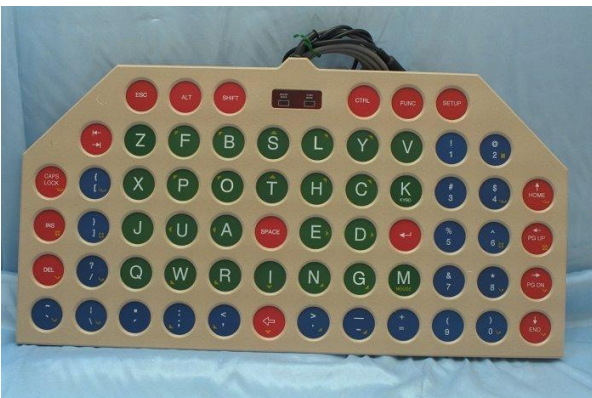
Alternative Keyboards



Examples of Alternative Keyboards for Direct Selection

Category	Description	Device Name
Expanded keyboards	Expanded keyboards have enlarged target areas, often programmable so that key size can be customized; useful for individuals with good range and poor resolution; also useful for individuals with limited cognitive or language skills or visual impairment	Big Keys LX, , Early Learning Keyboard, (Ablenet), Jumbo XL II Hi-Visibility Keyboard, HelpiKeys (Inclusive Technologies); Clevy Keyboard (Spectronics)
Contracted keyboards	Miniature, full-function keyboards, typically with membrane overlay; useful for individuals with limited range of motion and good resolution	The Magic Wand Keyboard (In Touch Systems)
Touch screens and touch tablets	Activated by either breaking a very thin light beam or by a capacitive array that detects the electrical charge on the finger; the electrode array used to detect where the finger or pointer is touching is transparent; the touch screen can be placed over the face of a monitor	Touch Windows for various size monitors (Magic Touch);
Special-purpose keyboards	Keyboards on special-purpose devices, such as augmentative communication and environmental control devices; available keys may be much more limited in number or may be specific in function compared with a standard keyboard	

Alternative Keyboards

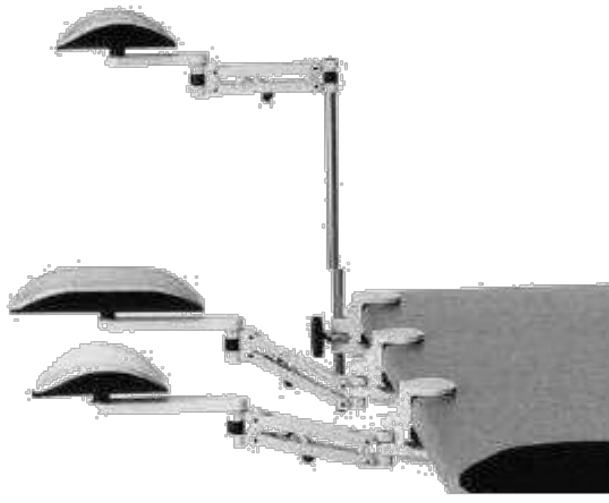


类别	说明	设备型号
Expanded keyboards 放大键盘	按键区域面积更大；大多支持可编程，按键尺寸可自定义； <u>适合肢体活动幅度尚可，但手部定位精度差的使用者</u> ；也适合认知/语言能力有限、视障人群	Big Keys LX、Early Learning Keyboard(Ablenet)、Jumbo XL II高可见键盘、Helpikeys、Clevy键盘
Contracted keyboards 压缩式迷你键盘	微型全功能键盘，大多带有薄膜按键面板； <u>适合肢体活动范围受限，但定位精度尚可的使用者</u>	The Magic Wand Keyboard
Touch screens and touch tablets 触摸屏与触控板	两种触发方式：阻断细光束；电容阵列检测手指电荷；检测手指/触控笔触碰位置的电极阵列是透明的；可以直接覆盖在显示器表面	Touch Windows多尺寸触控膜
Special-purpose keyboards 专用键盘	搭载在专项设备上的键盘，比如辅助沟通设备、环境控制设备； <u>按键数量更少，功能是定制化，和普通键盘不一样</u>	—

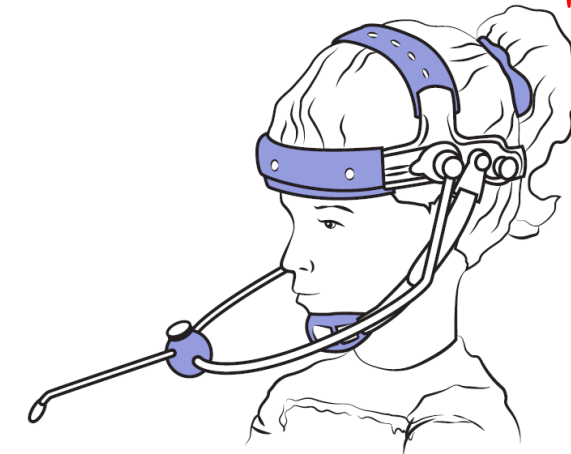
Device Name
Big Keys LX, , Early Learning Keyboard, (Ablenet), Jumbo XL II Hi-Visibility Keyboard, HelpiKeys (Inclusive Technologies); Clevy Keyboard (Spectronics)
The Magic Wand Keyboard (In Touch Systems)
Touch Windows for various size monitors (Magic Touch);

操控辅助配件. Control Enhancers

- 用户与键盘的摆放定位
- Positioning of user/keyboard
- Pointing devices • 定点输入设备
- Arm supports • 手臂支撑托架



A Mouth stick 口控棒



B 头控指点棒
Control enhancers. A, Mouth stick. B, Head pointer.

触摸屏

Touch Screens

- Optical: • 光学式触摸屏:
 - Touch interrupts a tiny light beam
 - 触摸动作会阻断细微光束
- Capacitive: • 电容式触摸屏:
 - Finger electrical capacitance modifies and is detected in an electric circuit

手指的电容改变电路参数，电路检测该变化实现触控识别



TouchWindow

触控膜产品TouchWindow

Pointing Devices

必须具备两项功能：

- 将光标移动到目标位置
- 将光标停留在目标位置足够长的时间

Required functions:

- Moving a cursor to a target
- Holding the cursor on the target long enough

轨迹球

Trackball Pro



EasyBall



Alternative Electronic Pointing Interfaces

Category	Description	Manufacturers
Keypad mouse 按键鼠标、	Mouse movement is replaced by keys that move the mouse cursor in horizontal, vertical, and diagonal directions. One or more keys perform the functions of the mouse button (click, double-click, drag). The ATEC Computer Switch Interface allows connection of up to five switches to a USB port. There is also a setting for mouse control. The user can use five switches to move the mouse up, down, left, right, and click.	Adaptation, Inc (ATEC Computer Switch Interface)
Trackball	Looks like an inverted mouse; a ball is mounted on a stationary base. Included on the base are one or more buttons that provide the functions of the standard mouse buttons. The base and hand remain stationary, and the fingers move the ball. Requires minimal range of motion and less eye-hand coordination.	Inclusive Technology, Trackman Marble Mouse and Wireless Trackball M570, Logitech, Roller Plus and Roller II Trackball (Traxsys Computer Products)
Continuous input joysticks	Joysticks (continuous input and switched) are used as direct selection interfaces for powered mobility. For computer use, movements are similar to wheelchair control; easy to relate cursor movement (direction, speed, and distance) to joystick movement.	Jouse2 (Compusult Limited); Roller Plus Joystick, Roller II Joystick, and EasiTrax (Traxsys Computer Products); JoyStick-C and JoyStick-C lite (Inclusive Technology); all manufacturers of power wheelchairs have their own joysticks, which are supplied with the wheelchair
Head-controlled mouse	An interface controlled through head movement; the user wears a sensor on the head, which is detected by a unit on the computer. Movement of the head is translated into cursor movement on the screen.	HeadMouse Nano (Origin Instruments Co.); TrackerPro (AbleNet); SmartNAV-4 (Inclusive Technology)
Camera mouse	A camera tracks a body feature (e.g., the tip of the nose), the movement of the feature being used to control an on-screen mouse cursor	CameraMouse (Boston College)
Switch mouse	A scanning strategy, controlled by one or two switches, is used to go over the on-screen targets	One Switch Mouse (Webbie); ScanBuddy (Applied Human Factors)
Light pointers and light sensors	These devices either emit a light beam that can be used to point to objects or as a control interface, or they receive light and provide an output when the light is reflected from an object or the light beam is interrupted.	(Laser light pointers can be used if the precautions shown in Box 8.1 are followed.)

Pointing Devices

Required functions:

- Moving a cursor to a target
- Holding the cursor on the target long enough



类别	说明	厂商
Keypad mouse 按键鼠标	用按键替代鼠标，按键控制光标上下左右斜向移动；一个或多个按键实现点击、双击、拖拽；ATEC接口最多接5个开关到USB，5个开关分别控制上、下、左、右、点击	Adaptivation
Trackball 轨迹球	相当于倒置鼠标；球体固定在底座上；底座自带鼠标按键；底座与手保持不动，只用手指拨动球体；肢体活动需求小，对手眼协调要求更低	Inclusive Technology、罗技M570无线轨迹球等
Continuous input joysticks 连续输入操纵杆	既用于电动轮椅直接操控，也用于电脑；光标方向、速度、位移和操纵杆动作一一对应，容易上手	Jouse2、EasiTrax，各大电动轮椅原厂操纵杆
Head-controlled mouse 头控鼠标	依靠头部运动控制；使用者头部佩戴传感器，电脑主机检测头部运动转化为光标位移	HeadMouse Nano、TrackerPro、SmartNAV-4
Camera mouse 摄像头鼠标	摄像头追踪身体特征（鼻尖等），身体运动转化屏幕光标运动	CameraMouse
Switch mouse 开关鼠标	依靠1-2个开关，采用扫描策略控制屏幕光标	One Switch Mouse、ScanBuddy
Light pointers and light sensors 光指点与光感设备	发射光束指向目标作为控制输入；或者接收物体反射光、光束被遮挡时输出控制信号	（使用激光指针需要遵守安全规范）

语音识别

Speech Recognition

- 离散语音识别：使用者每说完一个词必须停顿
- Discrete: require the user to pause between each word
- Continuous: more natural talking, without major pauses
 - 连续语音识别：说话更自然，不需要大幅停顿
- Speaker dependent: trained on a specific voice, recognition accuracy improves with training
 - 依赖说话人：针对特定人声训练；训练越多识别准确度越高
- Speaker independent: no training needed, vocabulary may be smaller
 - 不依赖说话人：无需训练，但可用词汇量通常更少。

眼动追踪系统 Eye-gaze Systems

视线的两种检测方式：

Gaze can be detected by:

- IR reflection: IR camera that detected the reflection of an IR beam by the retina • 红外反射法：红外摄像头检测红外光束经视网膜反射回来的光线
- Detection of pupil position wrt the eye, by a standard camera • 普通摄像头直接检测瞳孔相对于眼球的位置



人-辅助技术接口HTI的组成要素

Elements of HTIs



• 控制接口：

- **Control interface:**
 - 用户用来操作、控制辅助技术设备的硬件，
 - Hardware by which the user operates or control and AT device
 - Also referred to as **input device** • 也叫输入设备。
 - It generates inputs for the AT, in the input domain: 向辅助技术产生输入信号，输入分为两类：
 - Discrete inputs (e.g., keyboard) • 离散输入（例：键盘）
 - Continuous inputs (e.g., mouse) • 连续输入（例：鼠标）
- **Selection set: group of items available from which choices are made**
 - e.g., the keyboard elements, if typing in a computer 例：电脑打字时，键盘上全部按键就是选择集合。
 - e.g., controls for moving forward/backward and turning right/left, in a wheelchair 例：轮椅的前进后退、左右转向全部控制项。
 - Letters, words, symbols, icons, pictures • 选项形式：字母、词语、符号、图标、图片
 - Visual, auditory, tactile (e.g. Braille) • 反馈形式：视觉、听觉、触觉（例如盲文）
- **Selection method:** ③ • 选择方式：
 - 直接选择：用户直接在选择集合中指定目标选项
 - Direct selection: the user indicates the choice in the selection set directly
 - Indirect selection: intermediary steps are done for the selection • 间接选择：需要经过中间步骤，才能完成选择。



• 选择集合：可供用户从中做出选择的全部选项集合。

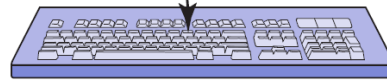
直接选择 vs 间接选择

Direct vs. Indirect Selection

手指/指点棒直接按下键盘按键

Direct Selection

Finger or
Pointer



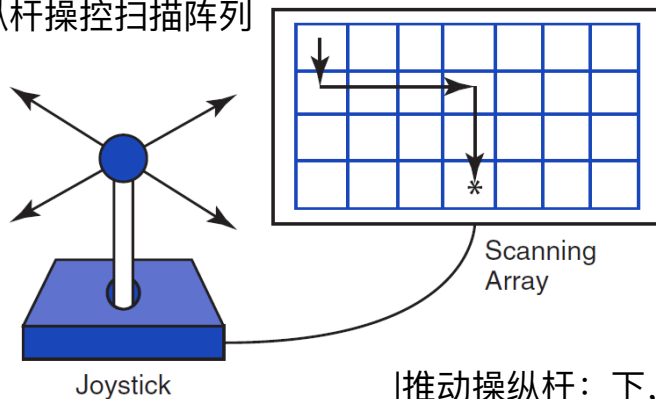
Keyboard

Input	Output
Press S 按下S键	S 输出字符S

The choice is done
directly in the
selection set

直接选择：直接在选项集合里完成选择。

图示：操纵杆操控扫描阵列



Joystick

|推动操纵杆：下，右，右，右，下|

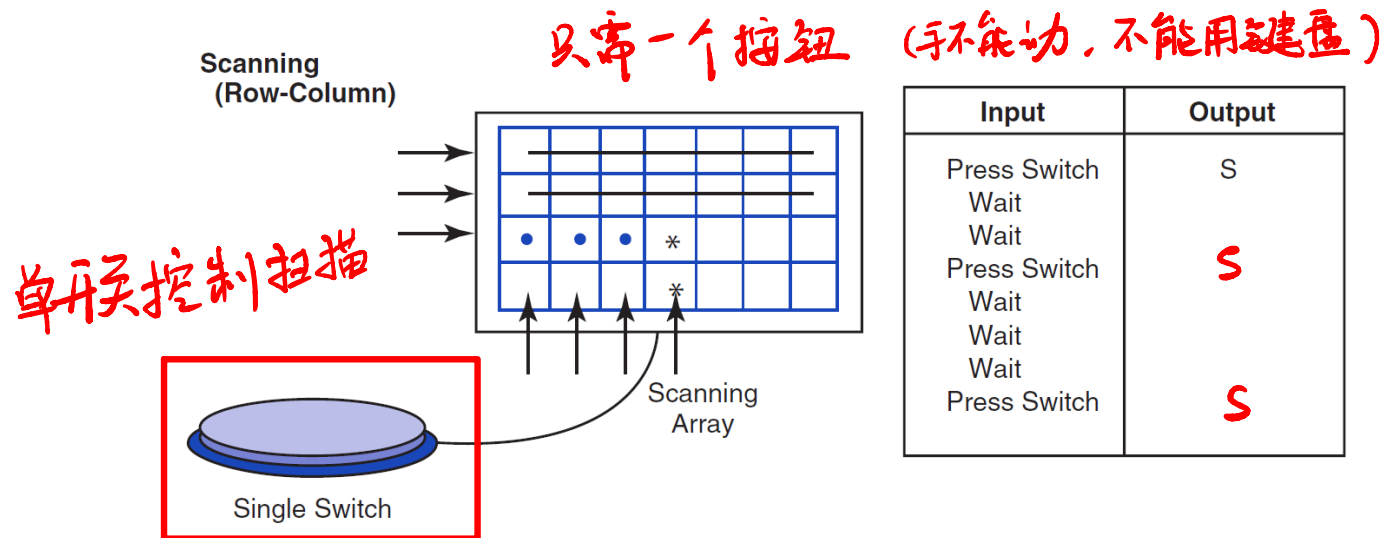
Input	Output
Move Joystick: Down Right Right Right Down	S

The choice is done
indirectly through a
sequence of steps

间接选择：要通过一连串步骤，才能完成选择。

☆ 间接选择：扫描选择法。 Indirect Selection: Scanning

- 选择集合展示在显示屏上。
 - The selection set is presented on a display
 - Each item in the selection set is sequentially lit or indicated by sound or speech
 - When the particular element that the user wishes to choose is presented, the user activates a control interface to select that item 当想要的选项被轮询到的时候，用户触发控制接口，完成选中。
 - Such control interface is typically a single switch, or an array of two or more switches
 - It requires very little motor control
- 集合内每一个选项会依次高亮，或者通过声音、语音播报提示。
- 这类控制接口一般是单个开关，或者2个及以上开关组成的阵列。
- 该方式对运动控制能力要求极低。



Scanning Techniques

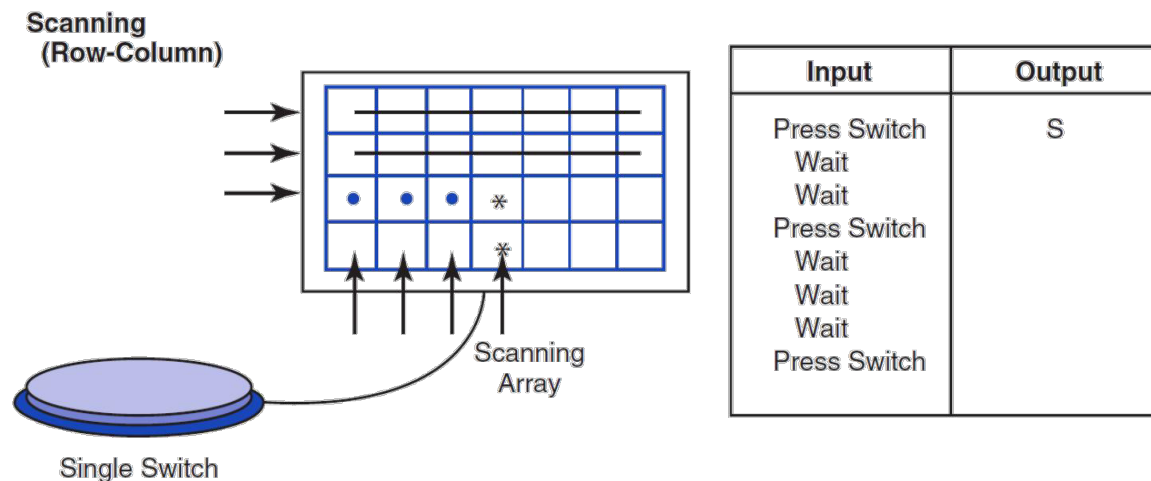
(1) 自动扫描：选项按顺序轮询；扫描速度可以根据用户反应快慢设置、调整。目标选项出现时，用户触发开关，停止扫描完成选择。

- **Automatic scanning:** items are presented in sequence. The rate of presentation (**scan rate**) can be set and adjusted according to how fast the user can respond. When the desired selection is presented, the user selects the choice by activating the control interface and stopping the scan.
- **Step scanning:** the user activates the control interface once for each item to advance through the choices in the selection set. When the user comes to the desired choice, there are two possibilities for selecting it:
 - 使用另外一个开关发出确认信号选中；
 - 使用确认延时：扫描停留在选项后等待一小段时间，时间到自动确认，不需要额外按键。
- **Inverse scanning:** the user initiates the scanning by activating and holding the control interface closed (e.g., keeping a switch pressed). As long as the control interface is held down, the items are scanned. When the desired choice appears, the user releases the control interface to make the selection.

(3) 反向扫描：用户按住开关，扫描就持续运行；看到目标选项，松开开关完成选择。

扫描排列格式 Scanning Formats

- 线性扫描：选项横向或纵向逐个轮询。
- 环形/旋转扫描：选项沿圆环逐个轮询。
- **Linear scanning:** items in the selection are presented horizontally or vertically, one by one
- **Circular/rotary scanning:** items in the selection are presented in a circle, one by one
- **Group-item scanning:** groups of items are scanned initially. The user select the group containing the desired element first. Then, the individual items are scanned and the user selects the item
 - 分组扫描：先扫描选项大组，用户先选中包含目标的分组；之后再扫描该组内的每一个子选项，再做选择。
 - e.g., row-column scanning



可用于操控辅助技术设备的人体部位。

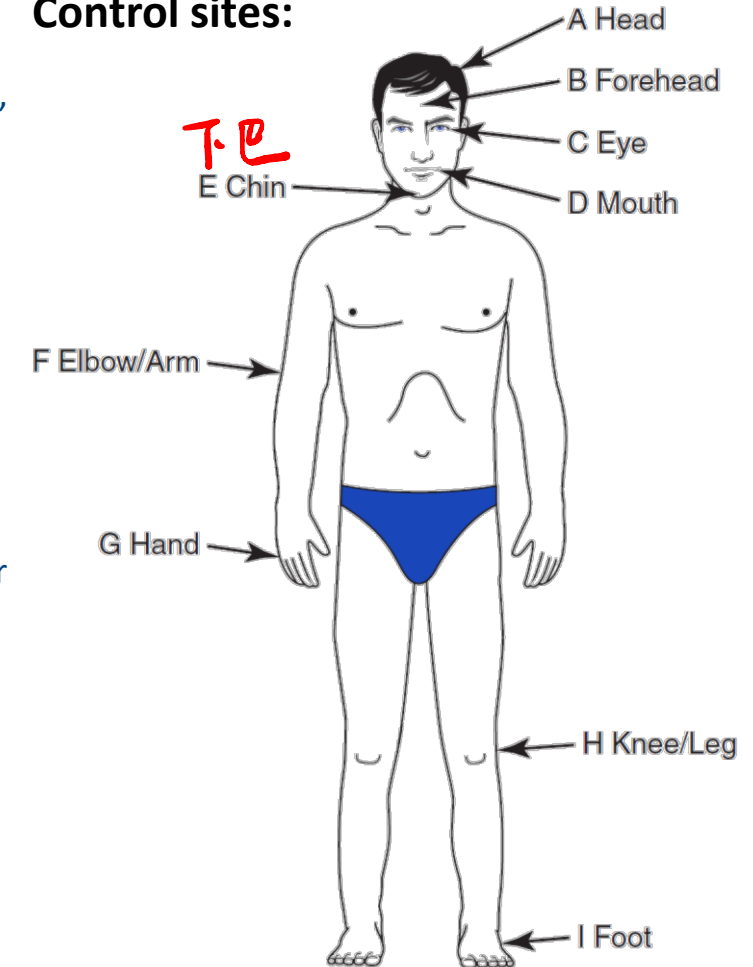
Anatomic Sites for Control of AT

- **hand or finger:** preferred because typically the most dexterous body parts.
 - The fingers can individually flex and extend or, together, perform a grasp and release movement.
 - The thumb can flex and extend, abduct and adduct, and oppose each of the fingers.
- **head:** relatively precise control using head movements such as tilting side to side, horizontal rotation, and tilting up and down. In practice, a combination of them (e.g. rotating left and tilting down)
- If both hand and head control are poor, then adapted HTIs, generally switches, can be used to detect movements of the shoulder, elbow, forearm, hand, or finger.
- **arm, leg:** less desirable for precise tasks because these represent naturally gross movements controlled by large muscle groups.
 - shoulder and scapular movements: elevation/depression, protraction/retraction, internal/external rotation, abduction/adduction, and flexion/extension.
 - elbow movements: flexion and extension
 - forearm movements: pronation (turning the palm down) and supination (turning the palm up)
 - wrist movements: flexion/extension, or from side to side (radial deviation or ulnar deviation).
- **foot:** for fine manipulative tasks, less desirable than the hand or head because visual monitoring can be difficult, and the foot is generally not as finely controlled as the hand. However, some individuals can develop fine control of their feet for typing.
- Control movements: raising and lowering the leg at the hip (e.g., flexion and extension), ankle plantar flexion (toes point down) or dorsiflexion (toes point up), and foot inversion or eversion (rotary movement, similar to pronation and supination).
- **mouth:** for switches based on respiration or phonation
 - respiratory air flow can be detected and used as a control site by sip (inhaling) or puff (exhaling) to access switches
 - phonation may produce sounds (including clicking or whistling) or speech.
- **eye:** gaze movements
- **tongue**

Each control site can perform a variety of movements or actions.

Each of these types of movements can be detected by an appropriate control interface.

Control sites:



Anatomic Sites for Control of AT

- /手指：首选，人体最灵巧部位。
 - 手指可以单独屈伸，也可以共同完成抓握、松开；
 - 拇指可以屈伸、外展、内收，与其余手指对捏。
- 头部：依靠头部动作可以实现较精准控制，动作包括左右侧倾、水平转头、抬头低头；实际经常组合动作，例如左转同时低头。
- 如果手、头控制能力都很差，则可以使用各类适配开关，检测肩、肘、前臂、手、手指的动作。
- 手臂、腿：不适合精细操作，属于大肌群主导的大幅度动作。
 - 肩与肩胛骨动作：上抬/下沉、前伸/后缩、内旋/外旋、外展/内收、屈伸。
 - 肘关节：弯曲、伸直
 - 前臂：旋前（手心朝下）、旋后（手心朝上）
 - 手腕：屈伸、左右摆动（桡偏、尺偏）。
- 脚：相比手和头，不适合精细操控；视线很难观察脚部，脚部精细控制能力弱。但部分人可以训练脚部完成打字。
- 脚部可执行动作：髋关节抬腿/落腿；踝关节跖屈（脚尖向下）、背屈（脚尖向上）；足内翻、外翻（旋转，类似前臂旋前旋后）。
- 嘴：用于呼吸发声类开关。
 - 检测呼吸气流：吸气、呼气来触发开关；
 - 发声：可以发出咔哒、口哨、说话作为控制信号。
- 眼睛：视线运动
- 舌头

每一个身体操控部位都可以完成多种动作，每一类动作都可以用对应的控制接口检测识别。

机械开关

Mechanical Switches

球形大按钮开关
Gumball



杯式小开关
Mini-Cup



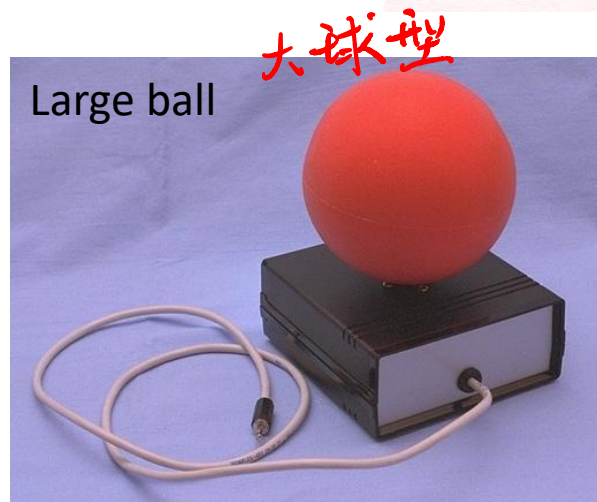
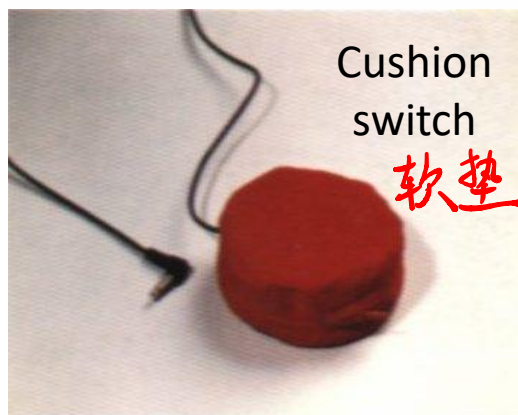
平板按压
Plate



气动
Pneumatic



角度可调
Adjustable angle
switch



Mechanical Switches

抓握

Grasp



吸吹气动

Sip and puff



脚踏

Pedal



手指

Finger switch



脚踏软垫

Mat



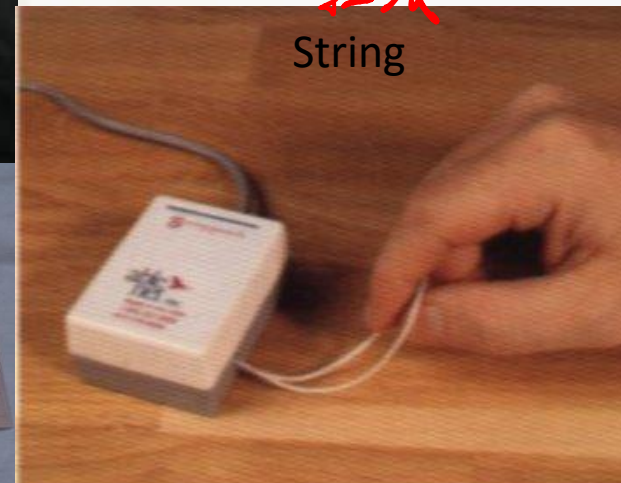
Soft touch

软触感



拉线

String



头控

Head switch



Other Switches

其它开关.

麦克风控制

Microphone



眨眼触发

EyeBlink



HTIs 间接接口总汇.

Indirect-selection HTIs

Examples of Single-Switch Control Interfaces		
Category	Description	Switch Name/Manufacturer
Mechanical switches	Activated by the application of a force; generic names of switches include paddle, plate, button, lever, membrane	Pal Pads, Taction Pads, and Flexible Switch (Adaptivation) Corp; Big Red Twist, Jelly Bean Twist, Ribbon Switch, Pillow Switch, Grasp Switch, Wobble Switch, several wireless switches and Buddy Button Switches (AbleNet, Inc.); Lolly, Thumb, Lever (long and short), Leaf (long and short), and Tread Switches (Zygo Industries); Flexible, Wobble, Picture Plate switch (AMDi); Egg, Egg w/finger mount, Lip, Button, Treadlite, micro-lite, Ultra Light, Tape Switches (Adaptive Switch Laboratories, Inc.), Cushion Grip, Ultimate (head control), Eyeblink Chin switch, and many other switches with various textured able surfaces, (Enabling Devices)
Electromagnetic switches	Activated by the receipt of electromagnetic energy such as light or radio waves	Fiber Optic Sensor (Adaptive Switch Laboratories, Inc.); SCATIR Minibeamer switch kit (AbleNet, Inc.) Infrared, sound, and touch switch (Words+; now AAC Works)
Electrical control switches	Activated by detection of electrical signals from the surface of the body	BrainFingers (AbleNet, Inc.), EMG Switch (Tinkerton, LLC), Twitch Switch, Switch Enabler kit (Enabling Devices)
Proximity switches	Activated by a movement close to the detector but without actual contact	Candy Corn switch (Ablenet) ASL 204M, 204-3 Pin, 208M, 208-3 Pin Proximity Switch, Honey Bee Switch (Adaptive Switch Laboratories, Inc.); Proximity Switches (AMDi)
Pneumatic switches	Activated by detection of respiratory air flow or pressure	Pneumatic Switch (Adaptivation Corp); Sip, Grip and Puff Switch (Toys for Special Children); ASL 308 Pneumatic Switch (Adaptive Switch Laboratories, Inc.); PRC Grip and puff switch (Enabling Devices), Pneumatic Switch (Zygo Industries)
Phonation switches	Activated by sound or speech	Voice Activated and Sound Activated Switches (Enabling Devices); Infrared/Sound and Voice Switches (Zygo Industries), Sound Activated switch (Enabling Devices)

Indirect-selection HTIs

类别	说明	开关名称&厂商
Mechanical switches 机械开关	依靠施加外力触发；包含桨式、平板、按钮、杠杆、薄膜开关等	Palm Pads、Big Red Twist、Jelly Bean Twist、枕头开关、抓握开关、摇摆开关、无线开关、唇控开关、下巴开关、头控开关等
Electromagnetic switches 电磁开关	接收光、无线电这类电磁信号触发	光纤传感器开关、红外声光开关套件
Electrical control switches 电信号开关	检测人体体表电信号触发	BrainFingers脑手指开关、EMG肌电开关、肌肉抽搐开关
Proximity switches 接近感应开关	物体靠近传感器就触发，不需要物理接触	Candy Corn接近开关、ASL系列接近开关
Pneumatic switches 气动开关	检测呼吸气流、气压触发	吸-吹开关、气动呼吸开关多款
Phonation switches 声控开关	依靠声音、语音触发	声音激活开关、语音触发开关

Summary

Questions ?

- 交互接口是辅助技术设备的核心部件，它让使用者能够操作设备。
- Interfaces are a key component of an AT device, since they give access to the device
- As such, they need to be suitable for the user's motor functioning
 - 接口必须适配使用者自身的运动能力。
- Indirect interfaces with scanning mechanisms are a very powerful way to provide access to AT to users with very few controllable motor actions
 - 带扫描机制的间接式接口，对仅有极少可控动作的使用者非常友好，让他们可以使用辅助技术。

➔ **they decouple the user's DOF from the device DOF**

间接接口把使用者的自由度，和设备的运行自由度做了解耦。